

BUILDING ROBUST NATURAL LANGUAGE PROCESSING SYSTEMS

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Abstract

Modern natural language processing (NLP) systems have achieved outstanding performance on benchmark datasets, in large part due to the stunning rise of deep learning. These research advances have led to great improvements in production systems for tasks like machine translation, speech recognition, and question answering. However, these NLP systems still often fail catastrophically when given inputs from different sources or inputs that have been adversarially perturbed. This lack of robustness exposes troubling gaps in current models' language understanding capabilities, and creates problems when NLP systems are deployed to real users.

In this thesis, I will argue that many different aspects of the current deep learning paradigm for building NLP systems can be significantly improved to ensure greater robustness. In the first half of this thesis, I will build models that are robust to adversarially chosen perturbations. State-of-the-art models that achieve high average accuracy make surprising errors on inputs that have been slightly perturbed without altering meaning, for example by replacing words with synonyms or inserting typos. For a single sentence, there is a combinatorially large set of possible word-level perturbations, so guaranteeing correctness on all perturbations of an input requires new techniques that can reason about this combinatorial space. I will present two methods for building NLP systems that are provably robust to perturbations. First, certifiably robust training creates robust models by minimizing an upper bound on the loss that the worst possible perturbation can induce. Second, robust encodings enforce invariance to perturbations through a carefully constructed encoding layer that can be reused across different tasks and combined with any model architecture. Our improvements in robustness are dramatic: certifiably robust training improves accuracy on examples with adversarially chosen word substitutions from 10% to 75% on the IMDB sentiment analysis dataset, while robust encodings improve accuracy on examples with adversarially chosen typos from 7% to 71% on average across six text classification datasets from the GLUE benchmark.

In the second half of the thesis, I will consider robustness failures that stem from the unrealistic narrowness of modern datasets. Datasets for tasks like question answering or paraphrase detection contain only a narrow slice of all valid inputs, so models trained on such datasets often learn to predict based on shallow heuristics. These heuristics generalize poorly to other similar, valid inputs. I will present methods for both constructing more challenging test data and collecting

training data that aids generalization. For the task of question answering, I will use adversarially constructed distracting sentences to reveal weaknesses in systems that standard in-distribution test data fails to uncover. In our adversarial setting, the accuracy of sixteen contemporaneous models on the SQuAD dataset drops from an average of 75% F1 score to 36%; on a current state-of-the-art model, accuracy drops from 92% F1 score to 61%. For pairwise classification tasks, I will show that active learning with neural sentence embedding models collects training data that greatly improves generalization to test data with realistic label imbalance, compared to standard training datasets collected heuristically. On a realistically imbalanced version of the Quora Question Pairs paraphrase detection dataset, our method improves average precision from 2% to 32%. Overall, this thesis shows that state-of-the-art deep learning models have serious robustness defects, but also argues that by modifying different parts of the standard deep learning paradigm, we can make significant progress towards building robust NLP systems.

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Chapter 1

Introduction

1.1 Recent successes in NLP

If standard benchmarks are to be believed, natural language processing (NLP) systems are now remarkably good at a wide range of language understanding tasks. The rise of deep learning, and in particular the success of unsupervised pre-training (Peters et al., 2018; Devlin et al., 2019), has led to impressive results across many NLP datasets. Consider the Stanford Question Answering Dataset (SQuAD), a dataset for the task of question answering (Rajpurkar et al., 2016). Given a paragraph from Wikipedia and a reading comprehension question, a system must find the word or phrase in the paragraph that answers the question. In the original SQuAD paper from June 2016, the best baseline system achieved only 51.0% F1 score.¹ Crowdworkers asked to answer these same questions got 91.2% F1 score, indicating significant room for improvement.² As of August 2020, the best system on the SQuAD leaderboard has 95.4% F1 score, surpassing the measured human-level accuracy.³

SQuAD is merely one NLP dataset that deep learning has “solved.” In the last few years, neural models have achieved near-human or super-human accuracy on many well-studied NLP datasets:

- Introduced in June 2018, SQuAD 2.0 was designed to be a more challenging successor to SQuAD, due to the inclusion of unanswerable questions that look similar to answerable ones (Rajpurkar et al., 2018). Between June 2018 and August 2020, the state-of-the-art improved from 66.3% F1 score to 93.0% F1 score, outperforming the human accuracy of 89.5% F1.⁴

¹The F1 score here is an evaluation metric defined in Rajpurkar et al. (2016) that gives partial credit for partially correct answers. This is distinct from the standard F1 score used for binary classification.

²Strictly speaking, the human and system scores for SQuAD are not directly comparable, as the human scores are computed against one fewer reference answer than the system scores. The evaluation metric gives credit for matching any reference answer, so this puts humans at a slight disadvantage compared to the systems. The numbers on SQuAD 2.0 reported below do not suffer from this problem.

³<https://rajpurkar.github.io/SQuAD-explorer/>

⁴<https://rajpurkar.github.io/SQuAD-explorer/>

- Introduced in April 2018, the General Language Understanding Evaluation (GLUE) benchmark is a compilation of ten datasets chosen to test a broad array of language understanding abilities (Wang et al., 2019b). Systems are evaluated on GLUE score, a macro average of task-specific evaluation metrics across the ten datasets. Between April 2018 and August 2020, the state-of-the-art improved from 60.3 GLUE score to 90.7, surpassing the human score of 87.1.⁵
- In response to the progress on GLUE, SuperGLUE was introduced in May 2019 as the successor to GLUE. It retains the format of GLUE but uses a more challenging suite of datasets (Wang et al., 2019a). Between May 2019 and August 2020, the state-of-the-art improved from 71.5 to 89.3, within half a point of the human accuracy of 89.8.⁶
- While the above results come primarily on datasets built for NLP research, NLP systems have also reached impressive accuracy on exams designed for (human) students. Introduced in April 2017, RACE (ReAding Comprehension dataset from Examinations) is a dataset of expert-written reading comprehension questions from English language standardized exams for Chinese middle and high school students (Lai et al., 2017). Between April 2017 and August 2020, the state-of-the-art improved from 44.1% accuracy to 90.9%. Crowdworkers on Amazon Mechanical Turk only got 73.3% accuracy on these questions (however, the authors estimate that an expert human could achieve 94.5%).⁷
- In September 2019, Clark et al. (2019b) reported 90.7% accuracy on text-based questions from a standardized science exam for American eighth grade students. In comparison, their initial system from 2014 could achieve only 36.4% accuracy, and their 2018 system could achieve 73.1% accuracy.

What should we make of these impressive results? It may be tempting to conclude from results on these *datasets* that current systems are as good as humans at many NLP *tasks*. However, this conclusion is not warranted.⁸ A dataset like SQuAD captures one particular distribution over documents and questions. The general task of question answering is much broader, encompassing

⁵ Baseline results come from the first version of the GLUE paper made public in April 2018 (<https://arxiv.org/abs/1804.07461v1>). State-of-the-art results are taken from the GLUE leaderboard (<https://gluebenchmark.com/leaderboard>). The best baseline reported on the GLUE leaderboard achieves 70.0 GLUE score and is dated February 2019.

⁶ <https://super.gluebenchmark.com/leaderboard>

⁷ http://www.qizhexie.com/data/RACE_leaderboard.html

⁸ Nevertheless, some media outlets have drawn such conclusions, including CNN (“Computers are getting better than humans at reading,” <https://money.cnn.com/2018/01/15/technology/reading-robot-alibaba-microsoft-stanford/index.html>) and USA Today (“Robots are better at reading than humans,” <https://www.usatoday.com/story/tech/news/2018/01/16/robots-better-reading-than-humans/1036420001/>). Many other media outlets did report on this same event in more tempered language, such as the Washington Post (“AI models beat humans at reading comprehension, but they’ve still got a ways to go,” https://www.washingtonpost.com/business/economy/ais-ability-to-read-hailed-as-historical-milestone-but-computers-arent-quite-there/2018/01/16/04638f2e-faf6-11e7-a46b-a3614530bd87_story.html).

a wide variety of documents that are understandable to humans and questions that humans could answer based on those documents. Since datasets are much narrower than the underlying task, impressive results on benchmark datasets can paint an overly optimistic picture of how well models work in general.

In fact, state-of-the-art systems achieve impressive benchmark results by specializing heavily to the datasets on which they are trained and evaluated—not by solving the general underlying task. In a standard benchmark setting, systems are built using provided training data, then evaluated on test data that was created in the same way as the training data. High accuracy on these benchmarks does not ensure that models can generalize to examples created in a different way, even though such generalization would be expected of a system that solves the underlying task. For example, we can ask how well a SQuAD model generalizes to TriviaQA (Joshi et al., 2017), another question answering dataset in the same format as SQuAD. While SQuAD was collected by asking crowdworkers to write questions about Wikipedia paragraphs, TriviaQA took questions written for trivia competitions and then matched them automatically with web documents. Yogatama et al. (2019) train a standard BERT-base model (Devlin et al., 2019) on SQuAD training data and achieve 86.5% F1 score on the SQuAD development set. However, this model only manages a paltry 35.6% F1 score on the TriviaQA development set.⁹ Despite its high accuracy on SQuAD, the model performs very poorly on other datasets in the same format.

1.2 The robustness problem

What is missing from the aforementioned SQuAD model, as well as current state-of-the-art NLP systems in general, is **robustness**. Since “robustness” has become a somewhat overloaded term, here we unpack exactly what we mean by robustness in this thesis. Sussman (2007) defines robust systems as “systems that have acceptable behavior over a larger class of situations than was anticipated by their designers.” In the context of machine learning, Schain (2015) defines robustness as “the sensitivity of a learning algorithm to discrepancies between the assumed model and reality.” Both definitions have a similar spirit, and seem like good properties for an NLP system to have, but are quite vague. What situations do designers of NLP systems anticipate? What assumptions do the learning algorithms they use typically make?

One important assumption on which most current work in NLP heavily relies is the *in-distribution* assumption—the assumption that the training data and test data are drawn from the same (or roughly the same) data generating distribution.¹⁰ The in-distribution setting is tailor-made for

⁹This gap is not due to TriviaQA being fundamentally much harder or noisier than SQuAD. Yogatama et al. (2019) also show that the same model architecture trained with a mixture of TriviaQA training data and other training data achieves 72.5% F1 score on TriviaQA. Training with SQuAD and TriviaQA alone achieves over 75% F1 score on TriviaQA, though the exact number is not provided. As of August 2020, the best system on the official TriviaQA leaderboard (<https://competitions.codalab.org/competitions/17208#results>) achieves 93.0% F1 on the Web verified test dataset, which is comparable to the evaluation set used by Yogatama et al. (2019).

¹⁰ For some datasets like MultiNLI (Williams et al., 2018), which is part of GLUE, the training and test distributions

the current dominant approach for building NLP systems, supervised learning (in particular, deep learning). In supervised learning, models are trained to produce the correct output on a set of training examples, with the hope that they will learn patterns that generalize to other test examples not seen during training. Supervised learning commonly assumes that training data and test data come from the same distribution, and standard approaches can be theoretically proven to generalize well in this in-distribution setting. In this thesis, we define robustness as the ability of a model to perform well on test data that differs in distribution from its training data—in other words, when the in-distribution assumption is violated.

1.3 Solving tasks requires handling worst-case examples

While we have established our focus on mismatches between training and test data, there is still a question of *how* the test data and training data differ. For instance, we could focus on robustness to domain shift, such as training on SQuAD and testing on TriviaQA. While this is an active area of research, we choose a different route. Much of the work in this thesis focuses on *adversarial robustness*, in which we evaluate on a worst-case (i.e., adversarially chosen) test distribution from a set of possible test distributions. As we will see throughout this thesis, models often perform very poorly on adversarially chosen test distributions that are seemingly very similar to the training distribution. The fact that accuracy drops precipitously when the data distribution changes in a seemingly small way makes adversarial robustness problems especially surprising and troubling.

In this section, we argue that adversarial robustness is closely related to the goal of solving language understanding tasks, rather than datasets. A system that does well on SQuAD but fails on other questions clearly has not solved the general task of question answering. To actually solve the task, a system would have to be able to produce reasonable answers when presented with any paragraph and question that is easy for a human well-versed in the language. Another way of phrasing this requirement is that *no adversary* should be able to find paragraphs and questions that stump the model but are easy for humans.

1.3.1 The Turing Test

In artificial intelligence (AI), the connection between true intelligent behavior (e.g., solving a broad task) and robustness to adversarial examples dates back many decades to Alan Turing’s famous Imitation Game, the Turing Test (Turing, 1950). Turing sought to define sufficient conditions under which a machine could be said to exhibit intelligent behavior. In the Turing Test, a human

differ, as some text sources were excluded from the training data and only present in the test data. Empirically, current systems generalize very well across these particular types of train-test mismatch, likely because the same data collection procedure was used to generate all examples, leading to persistent similarities despite the difference in text source. More minor differences in distribution are also somewhat common; for instance, SQuAD (Rajpurkar et al., 2016) uses different Wikipedia articles at training time and test time.

interrogator poses natural language questions to an agent—either a human or a machine—in order to determine whether the agent is in fact a human or machine. If a competent interrogator cannot distinguish the machine from a human, Turing argued that this demonstrates human-like intelligence on the part of the machine.¹¹

Crucially, Turing’s interrogator is *adversarial*: they are incentivized to ask precisely those questions that would reveal non-humanlike behavior on the part of the computer. Therefore, the machine can only succeed if it is prepared to answer any question that the interrogator could ask. The presence of the adversary does not imply that intelligence or language understanding is a fundamentally adversarial concept. Instead, the adversarial nature of this interrogation serves to ensure that the interrogator will ask a sufficiently challenging and diverse array of questions. Robustness in the face of such a strong adversary is strong evidence of intelligent behavior, since if the agent had some serious flaw, an intelligent adversary would be able to identify and expose it. Throughout this thesis, we will consider test distributions that are—to varying degrees—adversarially chosen to expose models’ weaknesses. Our ultimate goal is not to be antagonistic; on the contrary, we challenge systems so that we can identify and quantify their shortcomings and make measurable progress on building general language understanding systems.

1.3.2 Generalizing to the worst case in linguistics

The desire to account for worst-case instances also pervades the study of linguistics. This drive can be seen, for instance, in Richard Montague’s work on formal semantics (Montague, 1970, 1973). Montague developed an elegant theory of semantics based on the lambda calculus in which phrases of the same syntactic category could be assigned identical semantic types. These semantic types are sometimes non-obvious. For instance, Montague chose to treat adjectives as functions from properties to properties, even though many adjectives (e.g., “red”) are intersective (i.e., “red cars” denotes precisely the set of cars that are also red) and therefore could be thought to have much simpler extensions. However, Montague pointed to the existence of non-intersective adjectives like “big” (a “big flea” is not necessarily a big entity) and non-subsective adjectives like “alleged” (an “alleged intruder” need not be an intruder), and concluded that a proper account of all adjectives demanded a more general semantic type (Montague, 1970). Montague’s approach has been given the name “generalizing to the worst case” (Partee, 2007), as he purposefully constructed his theory so that it could uniformly handle these worst-case examples. We study worst-case examples with the same hope that they can shed light on ways we can improve our models of language.

¹¹We distinguish the original test proposed in Turing (1950) with a modern colloquial usage of the term “Turing Test” to refer to any evaluation in which a human must guess whether a generated output was produced by a human or a computer. While this is another type of imitation game, it lacks an adversarial interrogator, so passing such a test is a much weaker indication of intelligent behavior.

1.4 Motivating settings

We have so far argued that progress on language understanding requires consideration of worst-case examples, but have not specified the types of worst-case examples on which we plan to focus. While the Turing Test serves as theoretical inspiration, it does not provide any practical guidance on this question; it simply demands that all questions a human interrogator may come up with should be answered, an unrealistic standard for current models (Shieber, 2016).

One possible approach is to focus on narrowly-defined tasks for which a complete solution seems within reach, then scale up to broader tasks; we will consider this idea more in Chapter 7. However, most research on robustness does not alter the task definition, but instead studies how to defend against a constrained adversary that can only create certain kinds of hard examples, as opposed to the unconstrained adversary of the Turing Test. By restricting our purview to certain prescribed test distribution shifts, we can focus on solving specific aspects of the broader language understanding problem. Here we discuss several more constrained settings that provide motivation for studying adversarial robustness.

1.4.1 Systematic language understanding

To test whether a computer system understands natural language, it is natural to first ask what is known about natural language, and then determine whether the system knows this. We thus turn to the literature on linguistics and cognitive science for inspiration. In particular, one central concept in semantics is the Principle of **Compositionality**, which states that the meaning of a complex linguistic expression is a function of the meaning of its syntactic constituents and the way their meanings are composed. Compositionality is often invoked to explain the infinite productivity of language—from a finite set of lexicon entries and grammar rules, it is possible to generate an infinite set of acceptable sentences with understandable meanings. The related concept of **systematicity** provides us with a way to behaviorally measure whether a system understands language compositionally (Fodor and Pylyshyn, 1988). Humans language understanding is systematic, in that the ability to understand certain sentences is inherently tied to the ability to understand other sentences. If a person understands the sentence “*John likes the dog*,” they certainly also understand the sentence “*The dog likes John*.” There are simply no human beings in the world who would understand the former but be confused by the latter. Compositionality explains which sentences are related systematically with each other: two sentences must be systematically related if they involve the same syntactic constituents and same composition rules.

In Chapter 3, we will develop methods for achieving robustness to adversarial label-preserving word substitutions. These substitutions represent a very basic test of systematicity: if a system is correct on an input, and a word in that input is replaced with a word that the model is familiar with and is synonymous (or nearly synonymous) with the original word, it ought to continue predicting

the correct answer. A failure to do so would indicate a lack of systematicity. We will also discuss possible ways to extend this setting to consider compositional structure beyond simple word-for-word substitutions.

It is of course the case that compositionality and systematicity do not encompass all of the important things known about language. Moreover, in this thesis we will only test systematicity in a very restricted sense. Our aim is to start with the most elementary properties of language, and assess whether systems have mastered these basics. We believe this to be particularly important given the black-box nature of modern deep learning models, which makes it difficult to reason about their likely failure modes. In the past, standard statistical models had structures that made at least some of their flaws easy to predict. For example, bag-of-words models are insensitive to word order; n -gram language models are unable to recall long-range context; probabilistic context free grammars and conditional random fields make explicit conditional independence assumptions. However, neural networks pose a challenge for such *a priori* analysis of model shortcomings, as neural networks of sufficient complexity can famously approximate any function of their input. Judging from their impressive benchmark results, neural networks have likely managed to *approximately* learn about many linguistic phenomena, but their inscrutability should make us question whether they have mastered any of them. Adopting this mentality, we therefore start with the simplest tests of competence, and find that models indeed often fail these basic tests.

Also along these lines, in Chapter 5 we will construct examples that are reminiscent of minimal pairs, which are commonly used in linguistics to show how small changes in a sentence can give rise to shifts in meaning or acceptability. Again, these minimal pairs will purposefully be quite simplistic compared to the ones commonly studied in the linguistics literature, in order to find the simplest challenge that nonetheless confounds current NLP models.

1.4.2 Real-world adversaries

Not all adversaries are well-meaning theoretical interrogators; adversarial agents often have strong incentives to attempt to manipulate the predictions of real-world NLP systems. For example, spam detection systems must be able to identify spam in spite of spammers attempting to bypass these filters (Dalvi et al., 2004; Lowd and Meek, 2005). Search engine optimization rankings make use of textual features, and website creators have strong incentives to find ways to artificially increase their website’s predicted relevance to user queries. More recently, systems that flag hateful social media content for review must be robust to adversaries who wish to evade detection (Hosseini et al., 2017). Defending against these threats requires building systems that are robust to whatever alterations an attacker might apply to text in order to achieve the desired classifier behavior. In general, this space of possible alterations is very large, and may be very difficult even to define. Nonetheless, we can at least strive to make the adversaries’ jobs more difficult by making models robust to some types of common alterations, such as synonym replacements or typos.

1.4.3 Avoiding bias

It is often the case that available datasets do not exactly represent the data distribution of interest. One particularly problematic case is when the dataset is biased in some way against a particular demographic group, which often leads to model predictions that unfairly disadvantage members of that group. For example, naturally occurring data will often underrepresent minority groups, so systems can do well on average while having high error rates on these groups (Blodgett et al., 2016; Tatman, 2017; Buolamwini and Gebru, 2018; Sap et al., 2019). Datasets may also recapitulate stereotypes tied to historical injustices, which incentivizes models to learn these very stereotypes to achieve the best test accuracy (Zhao et al., 2017, 2018; Rudinger et al., 2018). Training data collected from current users of a system will likely be skewed towards users for which the system works well, leading to a feedback loop in which underserved users become increasingly underrepresented in the training data (Hashimoto et al., 2018). In many cases, avoiding these types of biases can be thought of as optimizing a worst-case objective in which an “adversary” may change the data distribution to include more minority group members, or include more examples that invert stereotypes. While this thesis does not focus on preventing bias against underrepresented minorities, we believe it is an important motivating case for work on robustness.

1.4.4 Realistic distribution shift

In general, deployed systems must be robust to many kinds of mismatches between the training data and test inputs. Many of these mismatches occur naturally, when user requests fall outside of the training distribution. For example, linguistic annotation systems, such as part-of-speech taggers or syntactic parsers, may be asked by users to annotate text from sources that were not present in the system’s training data. Generalizing to such cases is an important robustness challenge, although it can sometimes be unclear whether such generalization is possible if the training and test distributions are too different (Geiger et al., 2019). Work in domain adaptation usually assumes that some information about the domain of interest is provided, either in the form of a small labeled dataset or unlabeled data (Blitzer et al., 2006; Daumé III, 2007). In Chapter 6, we will consider another setting, extreme label imbalance, in which train-test mismatch naturally occurs. We will explore better methods of data collection, which allows us to circumvent the question of whether generalization is possible given the original training dataset alone.

Even when systems perform well on user queries on average, rare but catastrophic errors can lead to serious problems. In 2017, Facebook’s machine translation system mistakenly translated an Arabic Facebook post with the message “*Good morning*” into a Hebrew phrase that meant “*Attack them*” (Berger, 2017). As a result, the Israeli police arrested the man who made the post and detained him for several hours. The machine translation system was not robust to the uncommon way the person wrote “*Good morning*”, leading to very serious consequences. Deployed systems must avoid egregious errors like wrongly translating non-violent messages into violent ones, even on

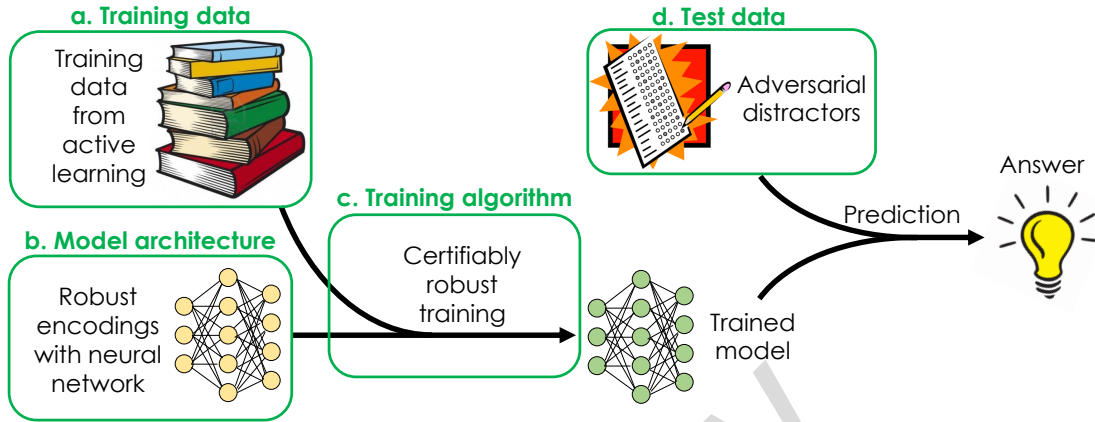


Figure 1.1: Work presented in this thesis on improving robustness, overlaid on a schematic depiction of a standard supervised deep learning pipeline for building NLP systems. In a standard supervised deep learning setting, training data (a) is used to learn parameters of a model architecture (b) via a training algorithm (c). This produces a trained model, which can take in test data (d) and output predicted answers. This thesis presents ways to improve these four components of this pipeline to yield more robust models. Certifiably robust training (Chapter 3) trains models to be robust to adversarial perturbations. Robust encodings (Chapter 4) are a model architecture component that also increases robustness to adversarial perturbations. Adversarial distracting sentences (Chapter 5) are one way to generate challenging test data to better evaluate robustness. Finally, active learning in imbalanced pairwise tasks (Chapter 6) collects training data that improves model generalization.

“worst-case” non-violent messages.

1.5 Building robust NLP systems

We have now seen many different robustness problems, and in particular adversarial robustness problems. Motivated by these problems, this thesis develops methods for building robust NLP systems. We focus on adversarial robustness problems in which reasonable systems should be expected to generalize, yet fail to in practice. Earlier, we noted that adversarially chosen test distributions that differ only slightly from the original training distribution can cause precipitous drops in accuracy. From the perspective of improving robustness, this closeness between the training and test distributions gives us some hope that robustness to these adversarial shifts is possible in principle, if we can find the right techniques.

While the robustness problems we consider are diverse, they share a central challenge: in order to ensure robustness, a system needs some way to compute, approximate, or bound the error incurred by worst-case inputs selected from a space of possibilities too large to enumerate. Standard neural NLP models have no way to reason about these worst-case scenarios. In this thesis, we show

how to modify various parts of the standard deep learning pipeline to guard against worst-case inputs, as depicted in Figure 1.1. In the following chapters, we will develop new training algorithms (Chapter 3), new model architectures (Chapter 4), and new dataset collection methods (Chapter 6), all of which address the key challenge of guarding against worst-case inputs while operating within the deep learning framework. In this way, we build robust NLP systems that still leverage and build on the tremendous progress of the past few years.

1.6 Outline

This thesis will describe four pieces of work that study different aspects of robustness in NLP. In each case, we will design adversarial tests of robustness that introduce test examples that systems should be expected to handle. Despite the simplicity of the challenges these tests introduce, state-of-the-art models will fare poorly. In the first half of the thesis, we will focus on robustness to adversarial perturbations, a setting that allows us to formally define a worst-case robustness goal. We will develop new training procedures and model architectures that can optimize for these goals. In the second half, we will examine cases where weaknesses of the dataset creation procedure itself creates problems. We will develop new strategies for constructing both evaluation and training datasets.

1.6.1 Adversarial perturbations

In Chapter 3 and Chapter 4, we will study robustness to adversarial perturbations. We will build models that are guaranteed to give the correct prediction when an adversary applies label-preserving perturbations to words in an input. Guaranteeing correctness is challenging because the adversary may simultaneously perturb many words in an input, leading to an exponentially large space of total perturbations. Due to this combinatorial explosion, modifying the training dataset alone does not ensure robustness, as it is computationally infeasible to include all possible transformations in the data. Instead, we will either change the training algorithm or the structure of the model to create models that can provably guarantee robustness to adversarial perturbations.

In Chapter 3, we focus on label-preserving word substitutions, such as lexical paraphrases. We train models that are provably robust to these word substitutions through a certifiably robust training procedure. In particular, we use interval bound propagation (IBP) to minimize a computationally tractable upper bound on the worst-case loss that any combination of word substitutions can introduce (Dvijotham et al., 2018). IBP had been previously used to guarantee robustness to pixel-level perturbations in computer vision; our work shows how to extend IBP to discrete input spaces and to neural network architectures commonly used in NLP, such as recurrent neural networks. This chapter is based on Jia et al. (2019).

In Chapter 4, we will guard against errors caused by adversarially chosen typos. Instead of relying on IBP, which makes assumptions about model architecture and scales poorly with network